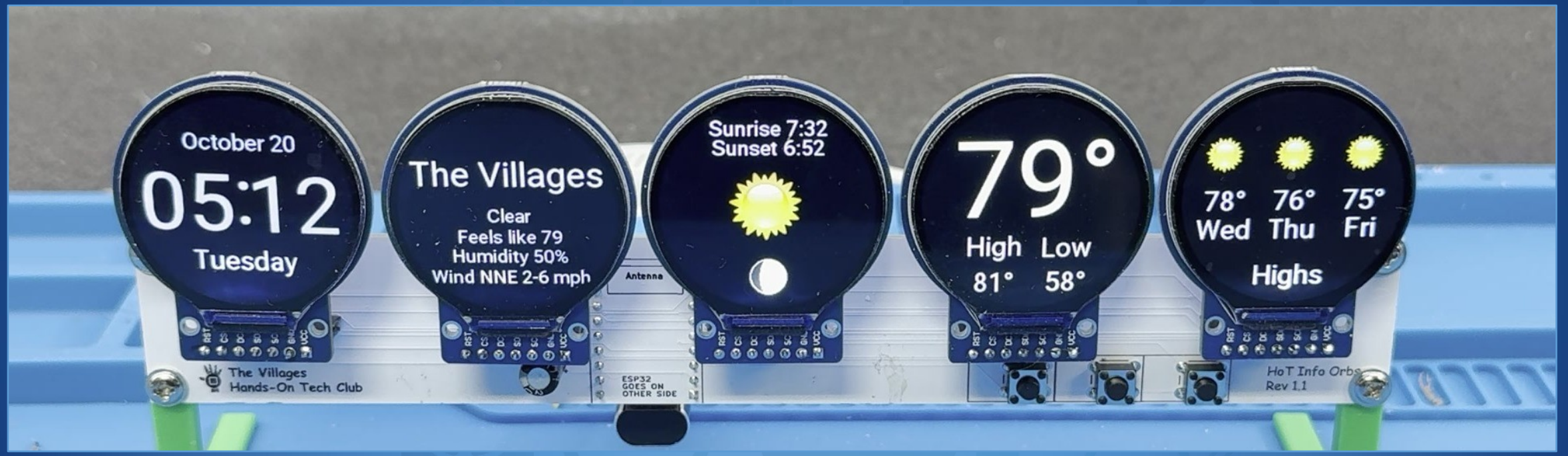
HOT Info Orbs

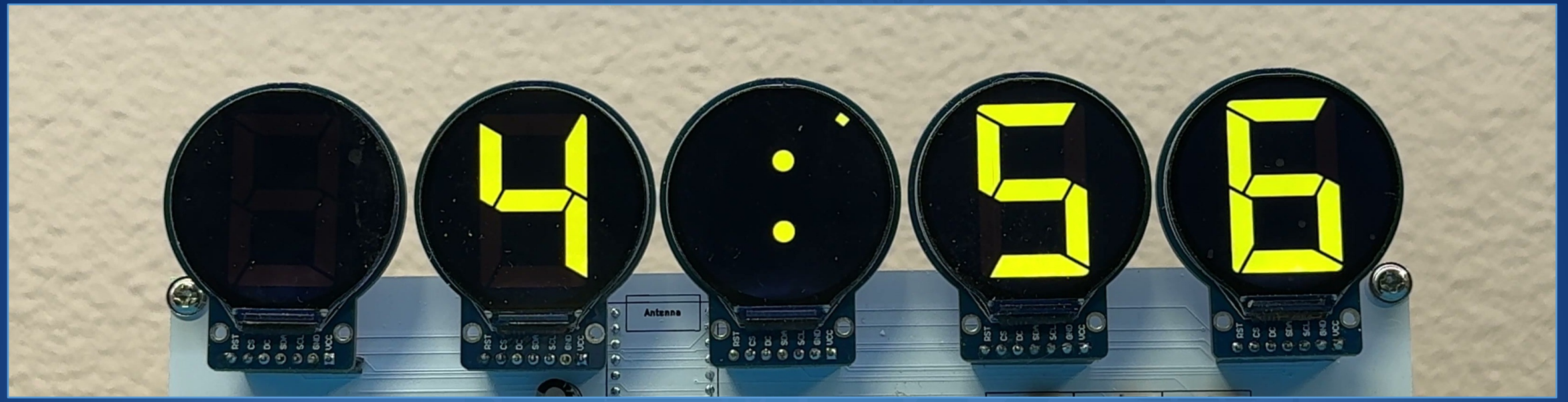
Five round screens that show what you want to know



Hands-On Tech, Small Computers group · October 27, 2026 · Scott Dube

A club build: a board designed by Yang and Scott, firmware and docs on GitHub

What it is



Five round screens

Color displays an inch and a quarter across, on one board. Each page uses all five.

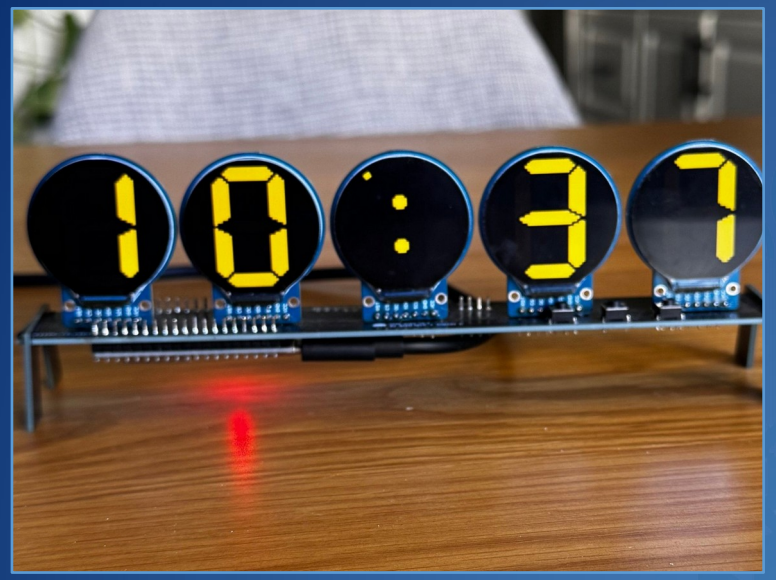
One small computer

A XIAO ESP32-S3 with Wi-Fi fetches the time, the weather, the markets and your own data.

Nothing to install

A USB cable for power, three buttons, and a settings page you open on your phone.

Where it came from



The original: Info Orbs

Brett Turner designed the Info Orbs and shared the plans and software for free.

Photo: Brett Turner, brett.tech



The kit: HackerBox 0129 “Orbital”

HackerBoxes, a monthly electronics kit, sent out its own version.

Image: HackerBoxes

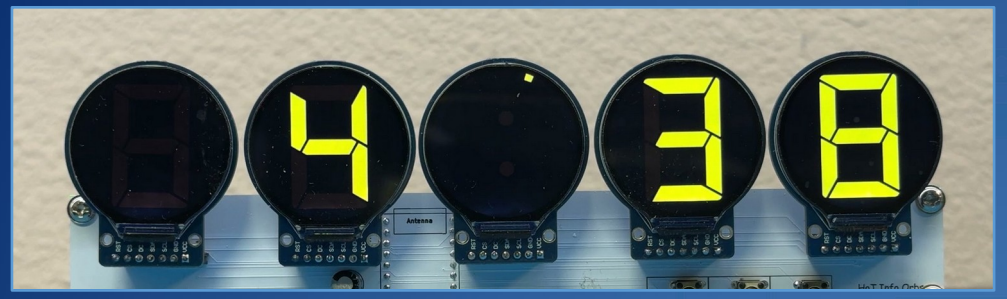


Built in the club

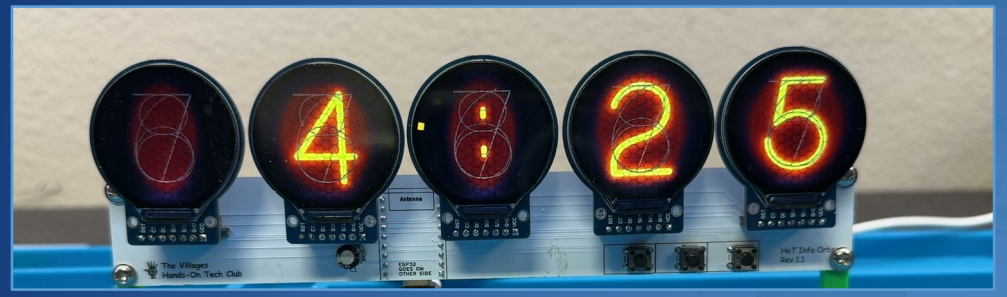
A couple of our members built that kit. We liked it enough to make our own: a new board, new features, a build anyone can follow.

Photo: Donald Champion

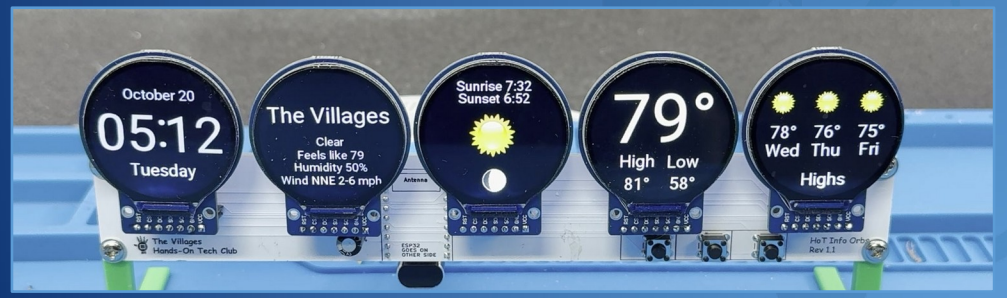
The pages: clock and weather



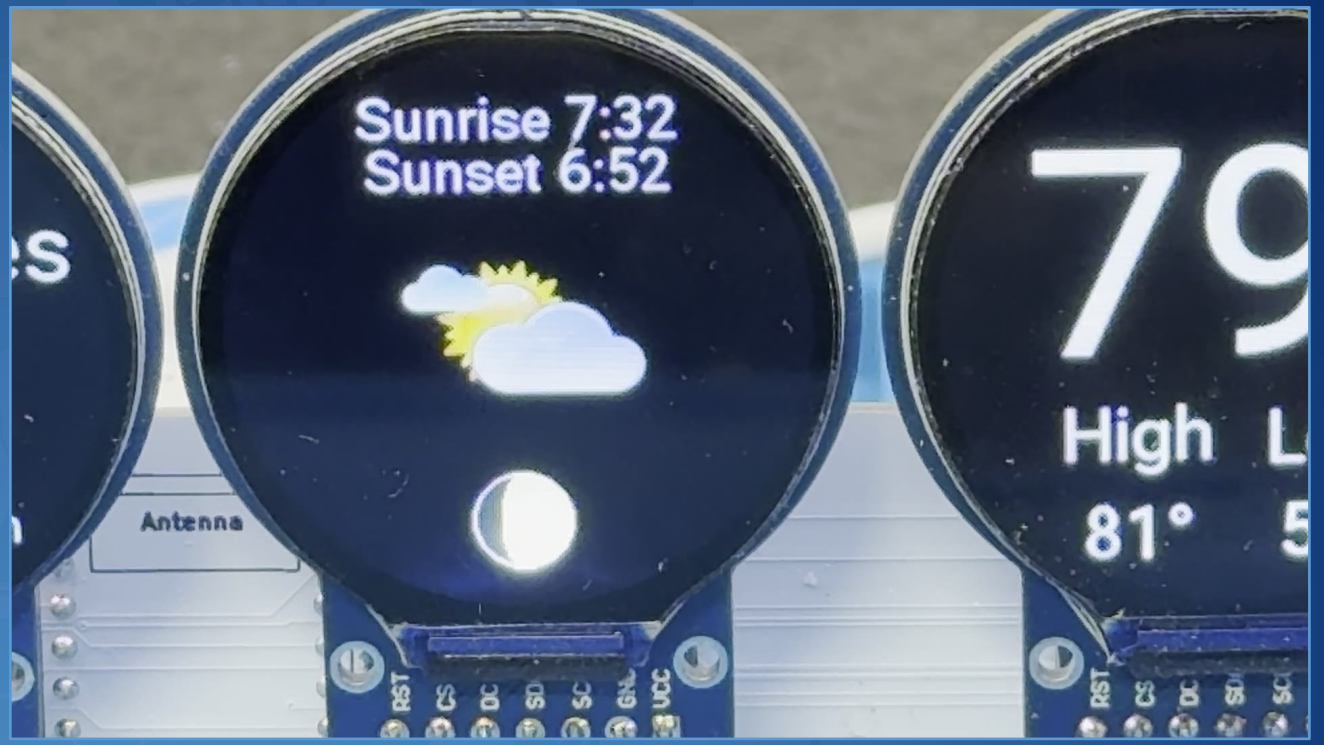
Clock, LED digits



Clock, Nixie face (middle button)



Weather: now, today, and the next three days



Sunrise, sunset and tonight's moon

The clock spreads across all five orbs: 12- or 24-hour, in any colors you like.

Weather for any town, or straight from your own Tempest station. Up to two stations.

The pages: stocks and your own data



Stocks

Five tickers, one per orb: price, change and the day's range. Stocks, funds or crypto.



MQTT

Five live values from Home Assistant, or anything else that speaks MQTT.

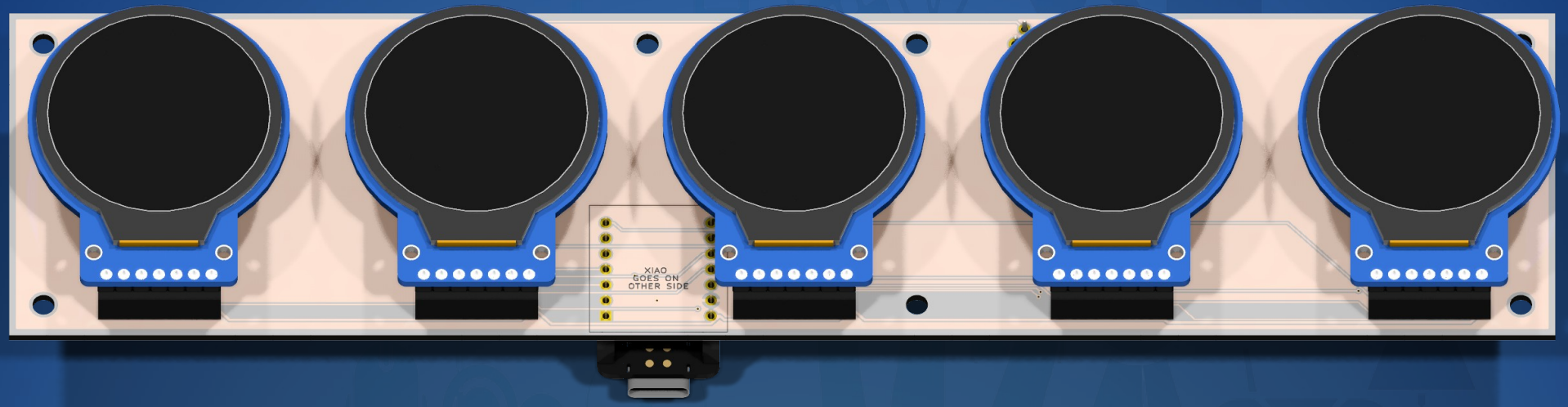


Web Data

A small JSON file from any web address, laid out the way you want.

Left and right buttons step through the pages. Leave it alone and it rotates on its own. The last two pages are ones you build yourself.

How it’s built



XIAO ESP32-S3 or S3 Plus
the brain, Wi-Fi built in, in a socket

Five 1.28 in displays
GC9A01 round color TFTs, socketed

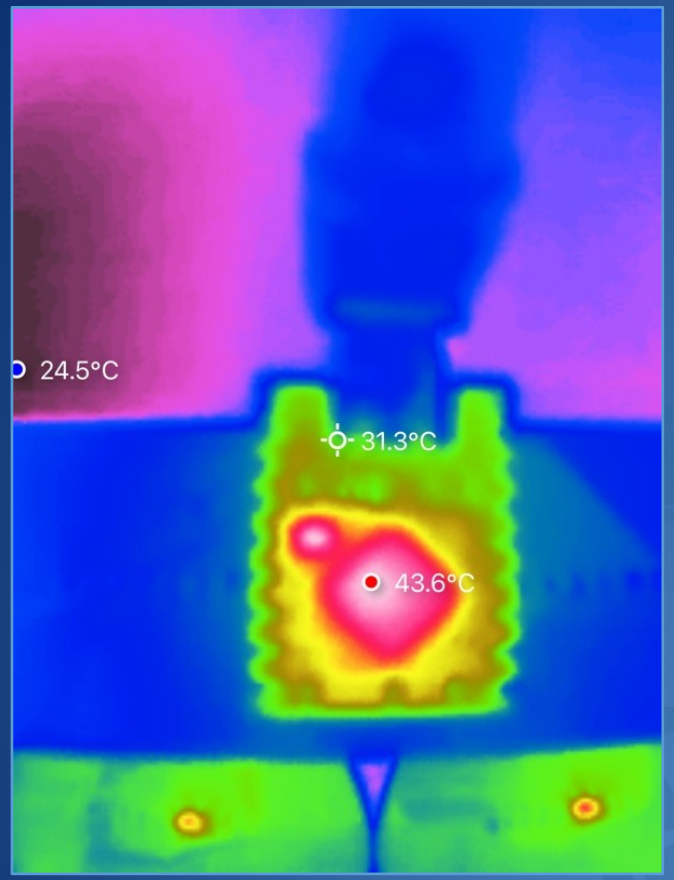
Three buttons
pages, clock face, wake

USB-C
power and the first firmware load

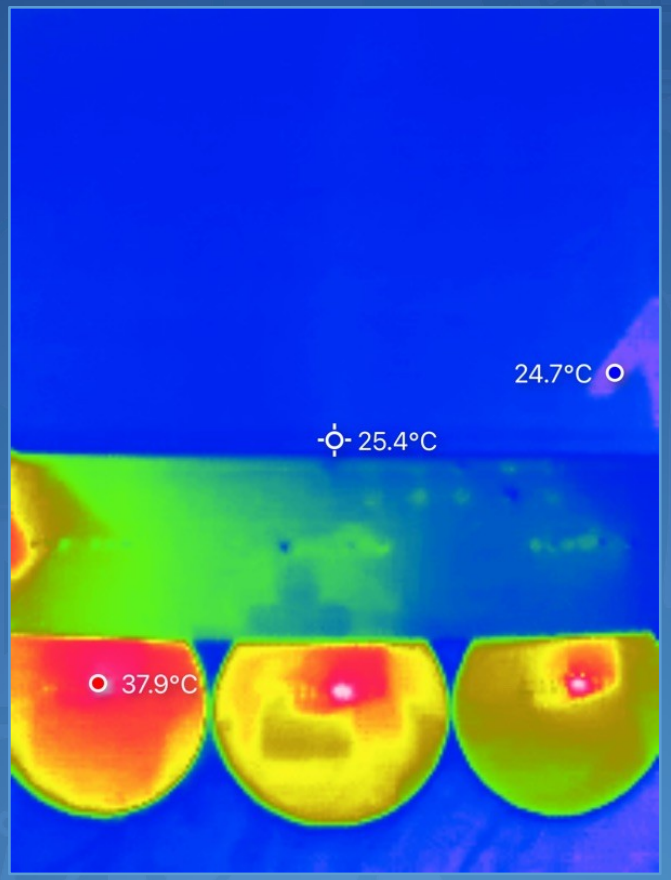
Under 2 W
1.7 W measured at 5 V

Board by Yang and Scott · made by JLCPCB · every file on GitHub

Why the displays run on 5 V



The ESP32 module, hottest point



The backs of three displays

✔ **43.6 °C**

our module, after hours of running

✔ **37.9 °C**

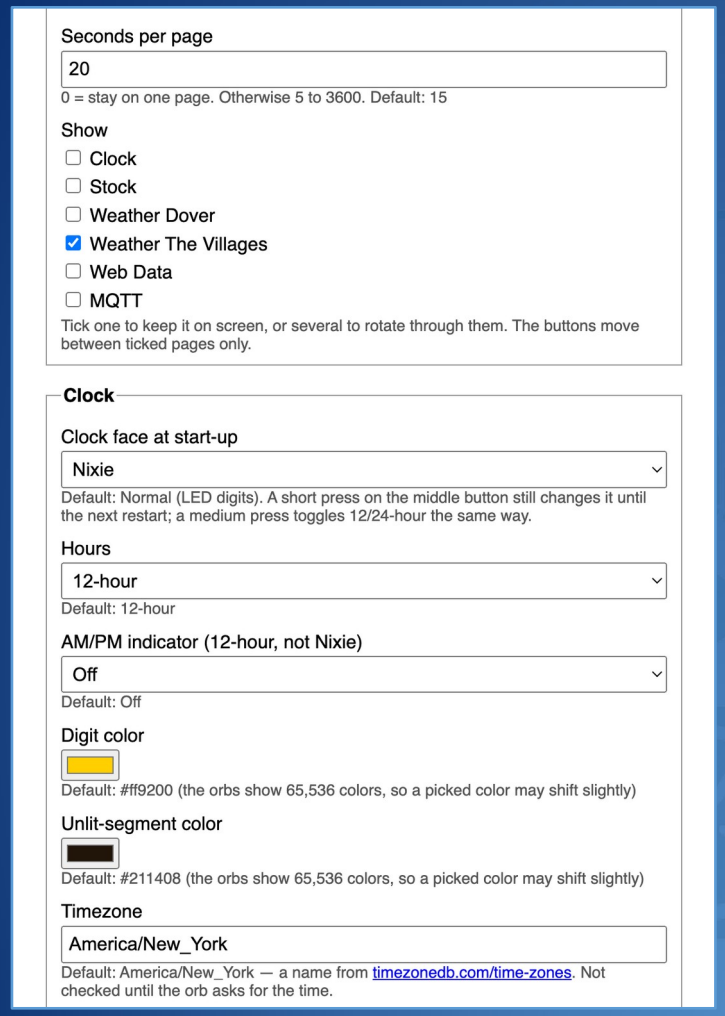
our display backs

✖ **82 °C**

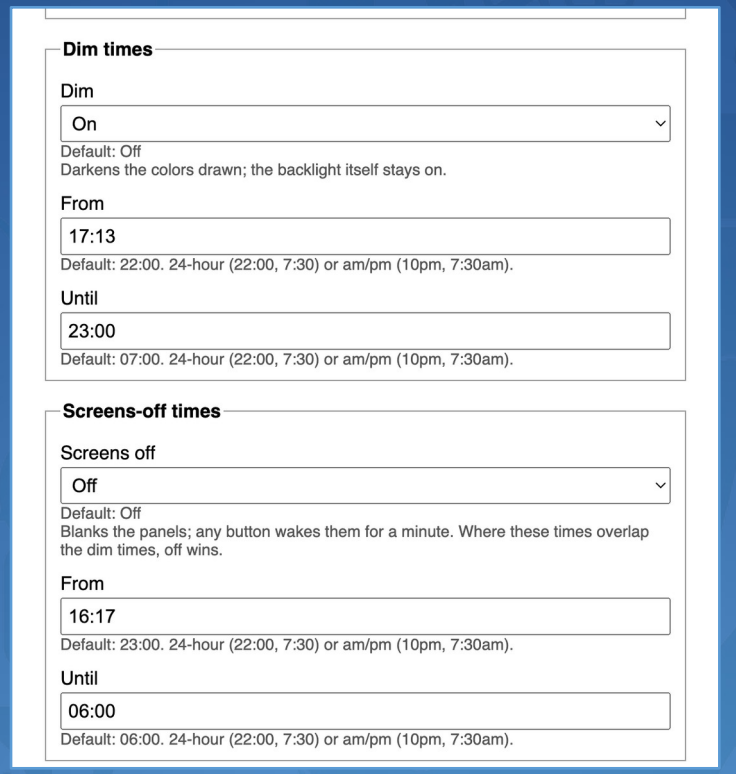
the kit’s module: “too hot to touch”, a builder’s report

On the kit every display ran off the ESP’s regulator. Here each display has its own 5 V feed, so they carry their own share and stay warm, not hot.

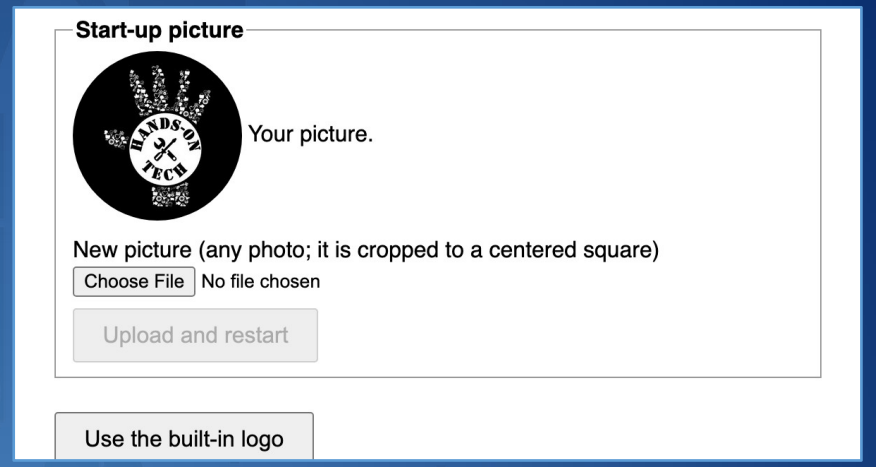
Setting it up from your phone



Pages and clock



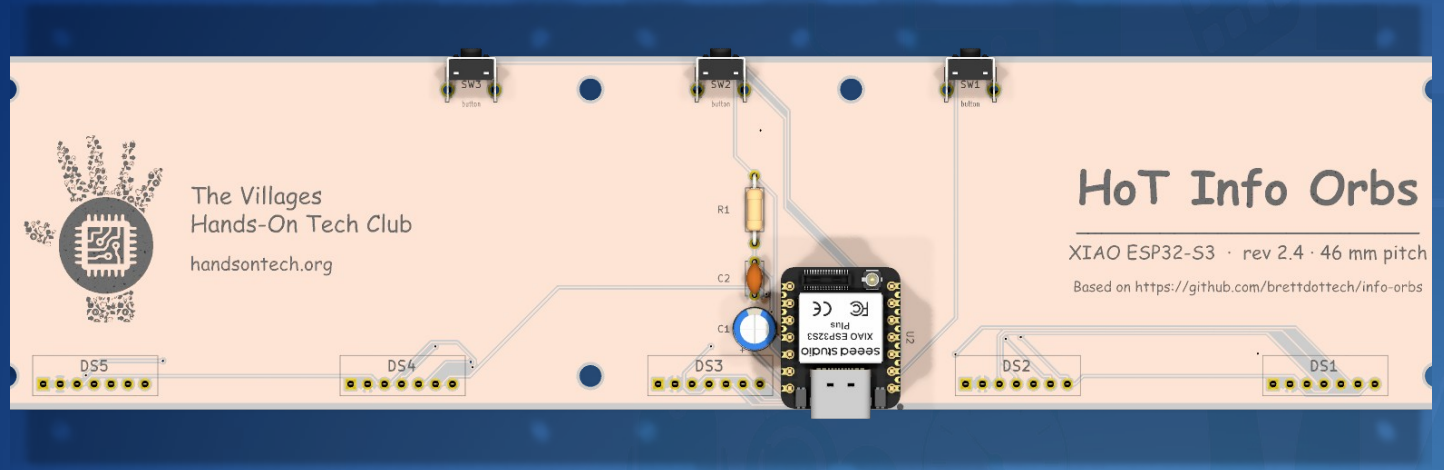
Dim and screens-off times



Your own start-up picture

- Tick the pages you want; untick the rest
- Your town, tickers, time zone, colors
- Dims in the evening, screens off at bedtime; any button wakes it
- Firmware loads once over USB, then updates over Wi-Fi from this page

What’s next





✓ Board rev 2.4: design locked

XIAO ESP32-S3 or S3 Plus, 46 mm display pitch to match the enclosure, buttons on top. This is the board the club builds.

○ Enclosure: concept, by Yang

“Winged Speedster”: brass rings around each screen, finned end caps, all 3D printed. Still a design.

The club build

1 Get the parts

Board, XIAO ESP32-S3 or S3 Plus, five displays, sockets, buttons, a USB-C cable.

2 Build night

Sockets and headers only: through-hole soldering, one evening.

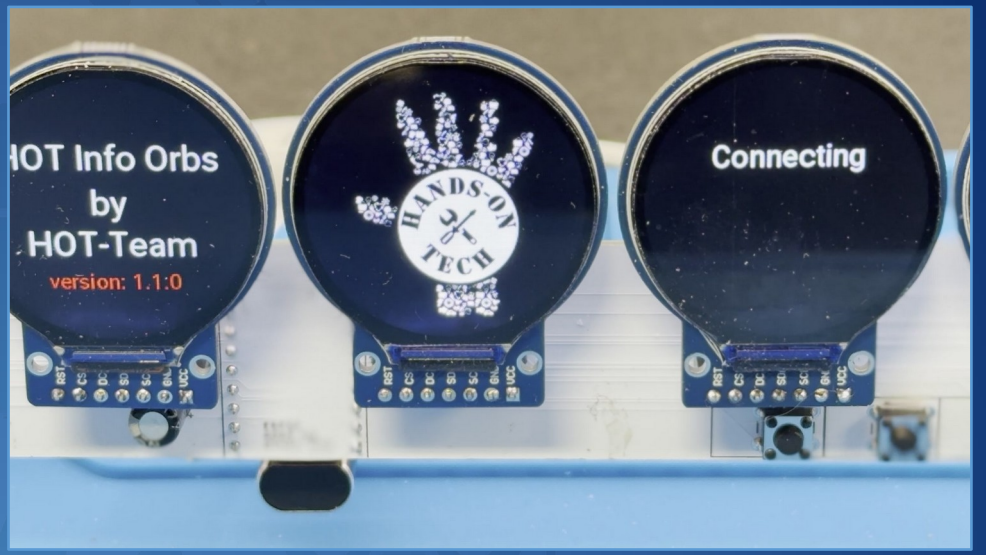
3 Load the firmware

Once, over USB. After that, updates from your phone.

4 Set it up

Pick your pages on the settings page. No code.

Starting in January. Say tonight if you want in.



Boots with the club logo

or any picture you upload. The module lifts out of its socket, so nothing is soldered that can’t be swapped.

Links and credits

Everything is on GitHub

github.com/scottdube/hot-info-orbs

Board files, firmware, build guide

Feature sheet

scottdube.github.io/hot-info-orbs/features.html

Thanks to

Brett Turner Info Orbs, the original

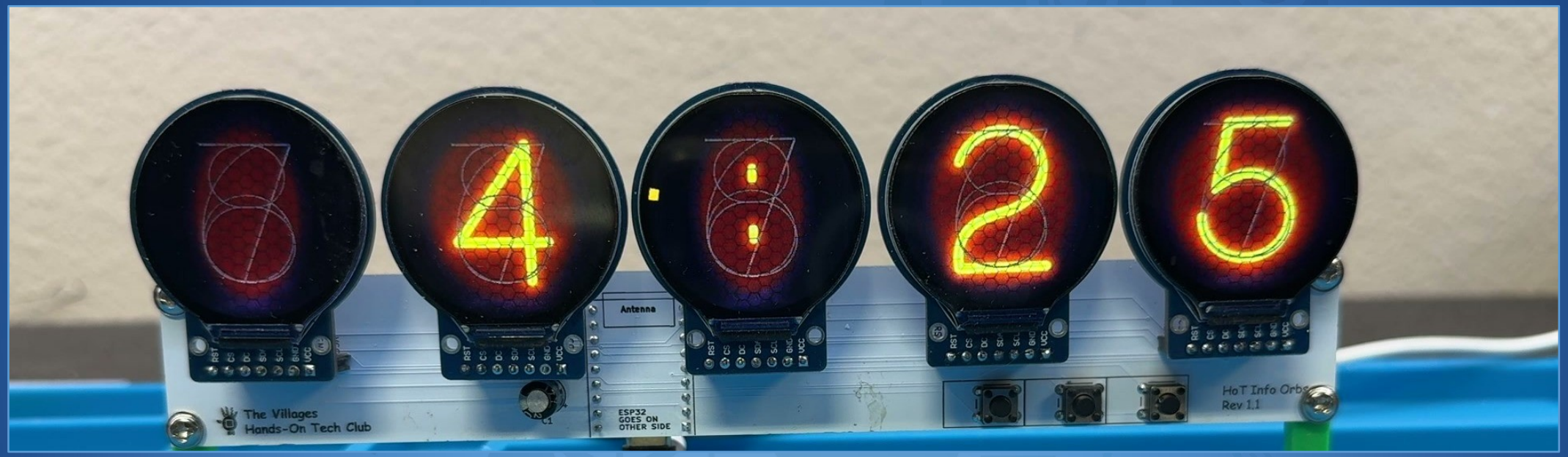
HackerBoxes #0129 Orbital

Donald Champion HackerBox build photos

Yang enclosure

Questions?

The orb is on the table. Press the buttons.



github.com/scottdube/hot-info-orbs